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Superconductor Project: Design a double-pancake solenoid wound with a HTS CICC for pulsed operation @ 15 T

NELAHTP - 2025/2026

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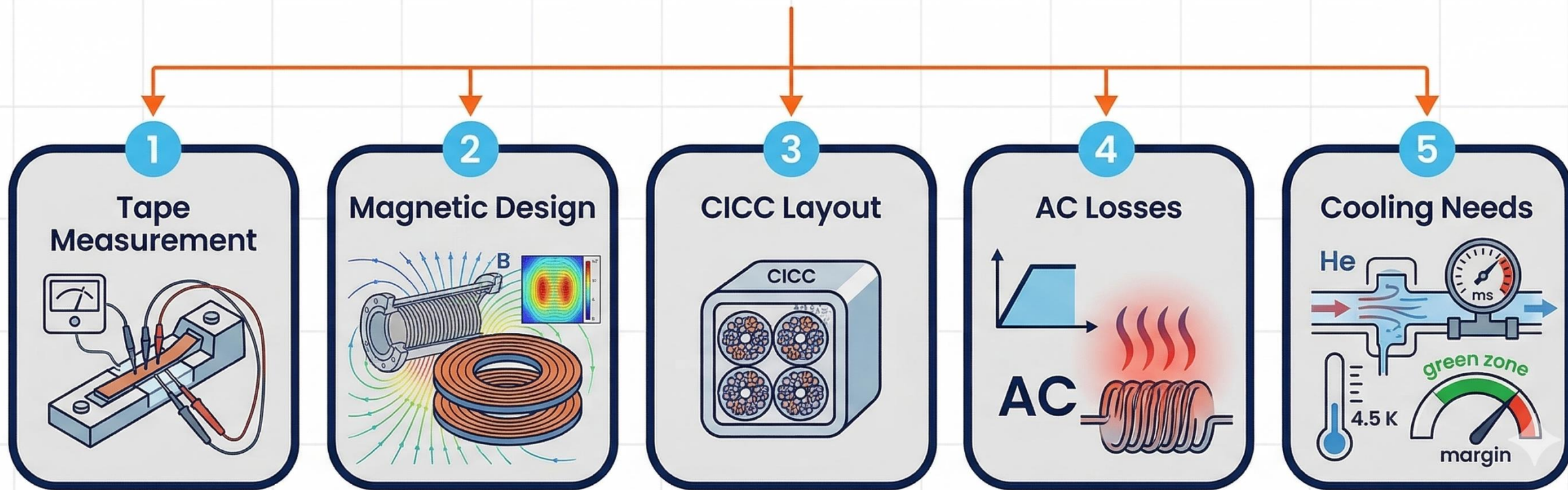
- ❖ SC: HTS REBCO tapes
- ❖ Stabilizer: Copper, RRR=100
- ❖ Jacket: Stainless steel
- ❖ Coolant: Helium, maximum available mass flow rate of 15 g/s, inlet temperature in the range [4.5 K – 50 K], inlet pressure in the range [5 bar – 20 bar];

The double pancake should be operated on the equatorial plane of a solenoid, wound using the same conductor, with inner diameter of 1 m and up to 1.5 m in height, sized to reach a maximum magnetic field on the conductor (including self field) of 15 T at the rated current of 50 kA. The current ramp rate in the solenoid is 0.2 kA/s.

Goal: Size the CICC and its cooling needs to guarantee that the temperature margin in operation is above 5 K



HTS Solenoid Conductor & Cooling Design Workflow



More in detail:

1. Measure HTS tape SC properties in LAB
2. Compute the B field map in the solenoid → compute the ampere-turn → compute the B field along your double pancake
3. For the measured strand props, identify the CICC layout (# of strands per slot, # of slots...)
4. For the operating scenario, compute AC losses
5. Assess the cooling needs: simulate the operating scenario to identify the minimum coolant mass flow rate that guarantees the rated temperature margin.

GOOD LUCK WITH YOUR WORK!



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